ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR DIGITAL BANKING TRANSFORMATION

DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

3. OPERATING MODEL OVERVIEW

3.1. Governance Model Type

✓ **Federated Governance Model**

- Central governance defines policies & standards
- Domain teams execute governance within business units

✓ **Recommended for Digital Banking because:**

- Multiple business domains
- Strong regulatory oversight
- Shared accountability
- Scalable

4. GOVERNANCE STRUCTURE

4.1. Organizational Layers

1. **Level 1: Board Risk Committee**

   Responsibilities:

   - Oversight
   - Risk appetite
   - Compliance review

2. **Level 2: Chief Data Officer (CDO)**

   Responsibilities:

   - Enterprise Data Strategy
   - Governance leadership
   - Regulatory alignment

3. **Level 3: Data Governance Council**

   Members:

   - CDO
   - CIO
   - CRO
   - COO
   - Head of Digital Banking
   - Head of Compliance

   Responsibilities:

   - Policy approval
   - Issue escalation
   - Data standards

4. **Level 4: Data Governance Office (DGO)**

   Responsibilities:

   - Framework execution
   - Data quality management
   - Metadata management
   - Stewardship management

5. **Level 5: Domain Data Owners**

   Domains:

   - **Customer Domain**
   - **Product Domain**
   - **Transaction Domain**
   - **Risk Domain**
   - **Finance Domain**

   Responsibilities:

   - Approve definitions
   - Data quality accountability
   - Issue resolution

6. **Level 6: Data Stewards**

   Responsibilities:

   - Data quality monitoring
   - Metadata maintenance
   - Business rules definition

7. **Level 7: IT Custodians**

   Responsibilities:

   - Security controls
   - Infrastructure
   - Backup & Recovery
   - Access management

5. RACI FRAMEWORK

Activity	Data Owner	Data Steward	IT Custodian	DGO
Approve business data definitions	A	R	C	C
Define and maintain DQ rules	A	R	C	C
Monitor DQ and coordinate remediation	A	R	C	C
Maintain business metadata	C	R	C	A
Approve business data access	A	C	C	I
Provision or revoke technical access	C	I	R/A*	I
Resolve domain data issues	A	R	C	C
Escalate cross-domain issues	C	C	C	R/A*

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

*R/A means Responsible and Accountable may be assigned to the same function for certain operational activities. If the matrix requires one Accountable role per activity, explicitly assign a single decision owner and separate approval from execution.

This matrix is a starting proposal. Adapt it to the bank's authority structure, particularly for sensitive data access, conflicts of interest, and information security.

6. CORE GOVERNANCE PROCESSES

6.1. Data Quality Management

6.1.1. Define Critical Data Elements (CDE)

Determination criteria:

- Regulatory/reporting impact
- Revenue/payment impact
- Customer experience impact
- Fraud/risk impact
- Operational dependency across domains

Critical Data Elements (CDE):

- Customer ID
- Account Number
- Mobile Number
- Transaction Amount
- Loan Outstanding

6.1.2. Data Quality Monitoring

KPI	Target
Completeness	>95%
Accuracy	>98%
Timeliness	>95%
Validity	>99%

6.1.3. Establish Data Quality Rules & Thresholds

6.1.4. Monitor via Dashboards

6.1.5. Issue Management Workflow

- Detect → Log → Assign → Resolve → Validate → Close

6.2. Metadata Management

1. Maintain enterprise data catalog
2. Define business glossary
3. Track data lineage (source → transformation → consumption)

6.3. Data Access & Security

1. Role-Based Access Control (RBAC)
2. Attribute-Based Access Control (ABAC) (optional advanced)
3. Access request & approval workflow
4. Periodic access review

6.4. Data Lifecycle Management

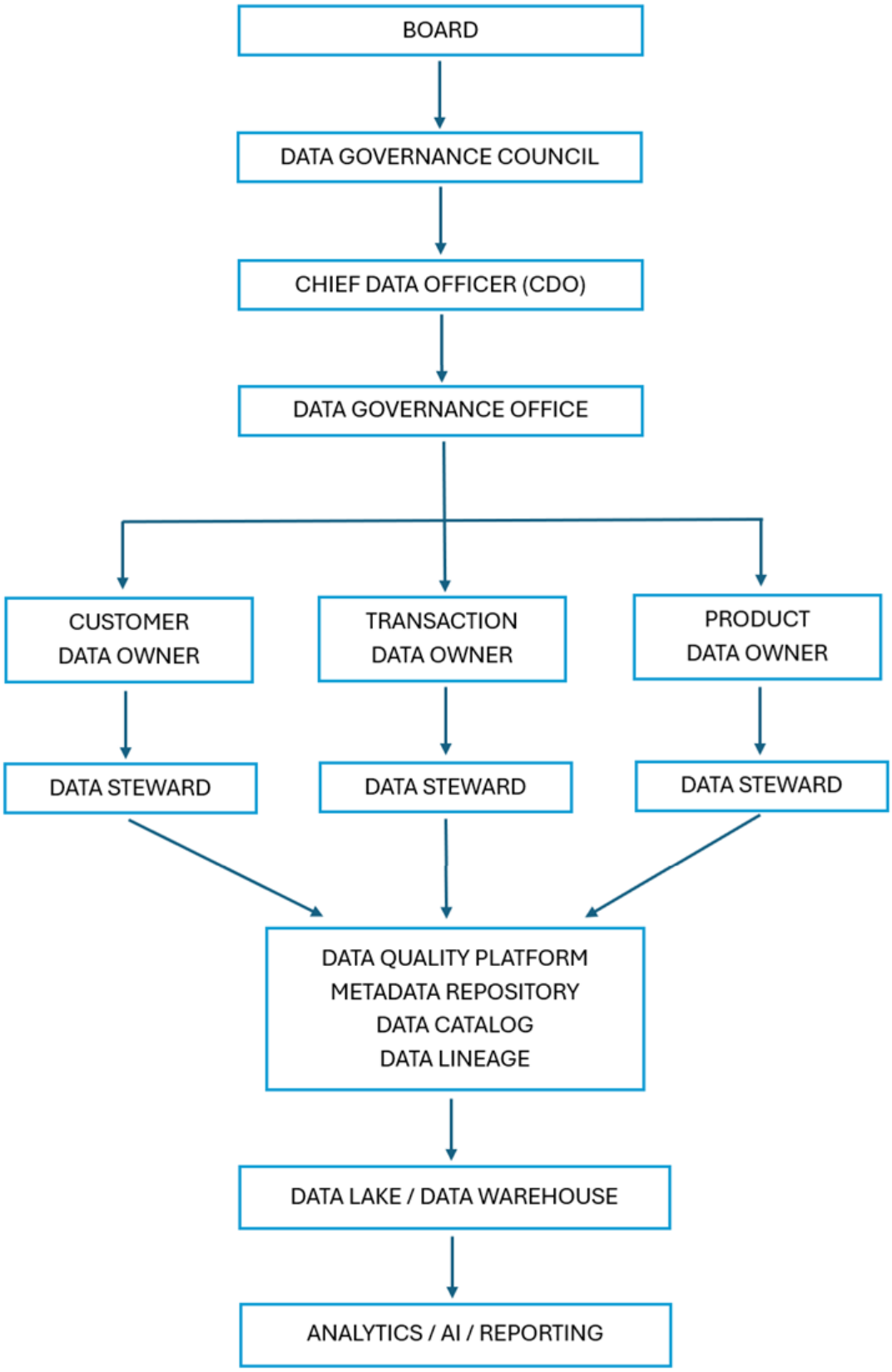
1. Define lifecycle stages:
    - Create → Store → Use → Archive → Delete
2. Retention policies based on regulatory requirements

6.5. Data Classification

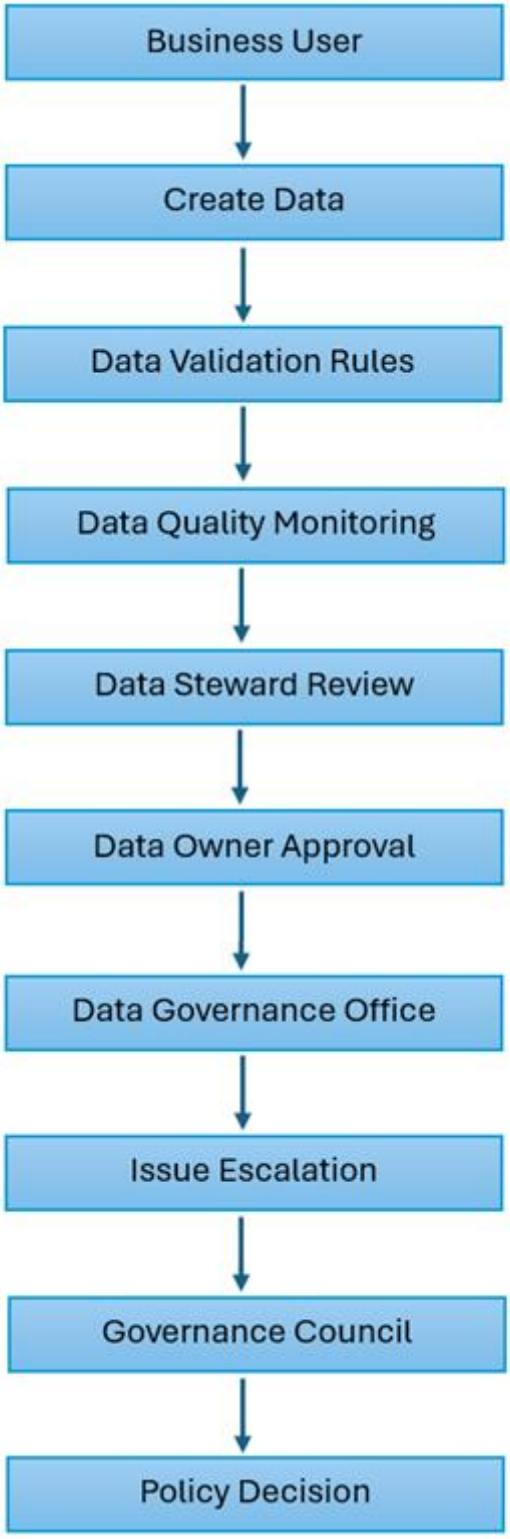
Define classification levels:

1. Public
2. Internal
3. Confidential
4. Restricted (PII / highly sensitive)

7. DATA GOVERNANCE ARCHITECTURE



8. GOVERNANCE FLOW DIAGRAM



9. GOVERNANCE ARTEFACTS

Mandatory Artefacts:

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

10. DATA GOVERNANCE FRAMEWORK COMPONENTS

10.1. Policy Layer

- Data Governance Policy
- Data Quality Policy
- Metadata Policy
- Data Security Policy

10.2. Process Layer

- Data Ownership
- Data Stewardship
- Data Issue Management
- Data Quality Monitoring

10.3. Technology Layer

- Data Catalog
- Metadata Repository
- DQ Tool
- MDM
- Data Lineage Tool

10.4. People Layer

- CDO
- DGO
- Owners
- Stewards
- Custodians

11. COMPLIANCE & RISK MANAGEMENT

1. Align with regulatory frameworks (e.g., GDPR principles)
2. Implement audit trails
3. Periodic compliance assessment
4. Risk identification and mitigation

12. KPIs & SUCCESS METRICS

KPI	Before	Target	Success Criteria
Data Governance Maturity	Level 1–2	Level 4	Managed & Measurable Governance
Data Quality Score	82%	≥98%	Trusted Enterprise Data
Metadata Coverage	10%	≥95%	Complete Data Documentation
Data Lineage Coverage	5%	≥90%	End-to-End Traceability
Regulatory Findings	8 Findings	0 Critical Findings	Regulatory Compliance
Duplicate Customer Records	15%	<2%	Single Customer View
Report Reconciliation Effort	3–5 Days	<1 Day	Faster Reporting
Data Issue Resolution SLA	20 Days	≤10 Days	Efficient Issue Management

13. IMPLEMENTATION ROADMAP

Phase 1: Assessment

Deliverables:

- Current State Assessment
- Gap Assessment
- Stakeholder Mapping

Phase 2: Framework Design

Deliverables:

- Governance Model
- Policies
- Standards

Phase 3: Technology Enablement

Deliverables:

- Data Catalog
- DQ Dashboard
- Metadata Repository

Phase 4: Pilot Implementation

Domains:

- Customer
- Transaction

Phase 5: Enterprise Rollout

Deliverables:

- Enterprise Governance
- KPI Dashboard
- Compliance Reporting
- Continuous Improvement

14. CHANGE MANAGEMENT & ADOPTION

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

15. ESCALATION FRAMEWORK

15.1. Data Issues Escalated

- Steward → Owner → Governance Council

15.2. SLA for Resolution Defined

SLA Table (Example):

Severity	Initial Response	Target Resolution
Critical	< 4 jam	< 24 jam
High	< 1 hari kerja	< 5 hari kerja
Medium	< 2 hari kerja	< 10 hari kerja
Low	< 5 hari kerja	< 30 hari kerja

16. REVIEW & CONTINUOUS IMPROVEMENT

1. Quarterly governance review
2. Annual TOM update
3. Continuous KPI monitoring

17. APPENDICES

Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

Appendix B: Sample Workflow

- Data Issue Management Workflow Diagram